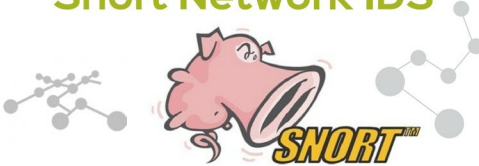


The Growing Popularity of the Snort Network IDS



Let's have a look at the premier, free, open source, network intrusion and detection system called Snort. It is an amazing tool that lives up to its billing. This tutorial walks you through the basics of Snort.

Snort is a very popular open source network intrusion detection system (IDS). It can be considered a packet sniffer and it helps in monitoring network traffic in real-time. In other words, it scrutinises each and every packet to see if there are any dangerous payloads. In addition, Snort can also be used to perform protocol analysis, content searching and matching. And it can be used to detect various types of attacks such as port scans, buffer overflows, etc. It is available on Windows, Linux, various UNIX as well as all major BSD operating systems. Snort doesn't require that you recompile your kernel or add any software or hardware to your existing distribution, but it does require that you have root privileges. It is intended to be used in the most classic sense of a network IDS. All it does is examine network traffic against a set of rules, and then alerts the systems administrators about suspicious network activity so that they may take appropriate action.

One can configure Snort in the following three different modes:

- Sniffer mode
- Packet logger mode
- Network intrusion detection mode

In the sniffer mode of operation, Snort will read network packets and just display them on the console. In packet logger mode, it will be able to log the packets to the disk. In network intrusion detection mode, Snort will monitor the network traffic and analyse it based on the rules defined by the user. It will then be able to take a specific action based on the outcome. Please note that Snort has an inbuilt real-time alerting capability.

As expected, network IDS are placed at certain points within the network so that the traffic to and from all devices on the network can be monitored. Once an attack or an abnormal situation is identified, an alert can be

triggered to the systems administrator to take corrective actions, as needed.

How to install and run Snort

Let us first understand how one can install Snort.

As a first step, execute the following command on your Linux terminal:

```
pswayan@pswayan-VirtualBox:~$ sudo apt-get install snort
```

Once the installation is complete, you can check how successful the installation has been by using the following command:

```
pswayan@pswayan-VirtualBox:~$ snort -version
```

Snort comes with a very detailed man page and readers should go through it for a thorough understanding of this IDS.

A few more programs are needed if you want to run Snort.

These are:

- Apache2 for the Web server
 - MySQL-server for the database
 - PHP5 for the server-based script
 - PHP5-MySQL
 - PHP5-gd for graphics handling
 - PEAR (PHP Extension and Application Repository)
- And we can use *apt-get* to install all the above programs as shown by the following commands:

```
pswayan@pswayan-VirtualBox:~$ apt-get install apache2
pswayan@pswayan-VirtualBox:~$ apt-get install mysql-server
pswayan@pswayan-VirtualBox:~$ apt-get install php5
pswayan@pswayan-VirtualBox:~$ apt-get install php5-mysql
pswayan@pswayan-VirtualBox:~$ apt-get install php5-gd
```